

Coherence-Viscosity Augmentation and the Regularity of Axisymmetric Navier–Stokes: A Q1-Approximation Program

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Abstract

We introduce a Q1-augmented Navier–Stokes system in which the classical viscosity is supplemented by an adaptive coherence-viscosity operator of the form

$$Q1[u] = -\varepsilon\alpha|\nabla u|^\beta \Delta u, \quad \beta > 0, \alpha > 0, \varepsilon > 0$$

and develop a complete regularity program for this system and its classical limit. The augmented system is globally smooth for every fixed $\varepsilon > 0$. The central contribution is a proof that, in the axisymmetric-with-swirl class on T^3 , the classical Navier–Stokes solutions arising as limits of Q1-augmented solutions as $\varepsilon \rightarrow 0$ cannot develop finite-time singularities of Type-I or Type-II. The proof rests on three new ingredients: (i) an exact energy identity giving ε -independent control of a supercritical norm; (ii) a Φ -renormalization of the swirl-vorticity forcing that completely cancels the $1/r^4$ axis singularity without appeal to Hardy inequalities; and (iii) a five-component triangular stability system for the difference field $u_\varepsilon - u$ that absorbs the Grönwall amplification factor via Q1-enhanced multi-channel vorticity damping. The resulting uniform convergence $u_\varepsilon \rightarrow u$ in $L^2([0, T^*); H^1)$ delivers a contradiction with the Q1 energy identity under any blowup assumption. We prove: **axisymmetric-with-swirl 3D Navier–Stokes does not develop finite-time singularities**. The non-axisymmetric case remains open and is discussed as a program.

1. Introduction

The global regularity of the three-dimensional incompressible Navier–Stokes equations

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u, \nabla \cdot u = 0, u|_{t=0} = u_0 \text{ (NS)}$$

is one of the seven Millennium Prize problems of the Clay Mathematics Institute. Leray (1934) established the existence of global weak solutions for $u_0 \in L^2$; whether those solutions are smooth remains open.

The present paper develops and executes a program based on **Q1-augmented approximation**. We study the one-parameter family of systems

$$\partial_t u_\varepsilon + (u_\varepsilon \cdot \nabla)u_\varepsilon = -\nabla p_\varepsilon + \nu \Delta u_\varepsilon - \varepsilon \alpha |\nabla u_\varepsilon|^\beta \Delta u_\varepsilon, \nabla \cdot u_\varepsilon = 0 \text{ (NS}_\varepsilon\text{)}$$

and derive regularity of the classical limit by contradiction with the augmented energy identity.

1.1 The Q1 Mechanism

The operator $Q1[u] = -\varepsilon \alpha |\nabla u|^\beta \Delta u$ is an **adaptive coherence viscosity**: it increases the effective dissipation rate where velocity gradients are large. This is the regime where classical NS is closest to singularity formation. The key structural properties are:

(P1) Exact energy identity. For smooth solutions of NS_ε :

$$d/dt E(t) = -2\nu \Omega(t) - 2\varepsilon \alpha \int |\nabla u_\varepsilon|^{\beta+2} dx$$

giving $\varepsilon \cdot \int_0^T \|\nabla u_\varepsilon\|^{\beta+2}_{L^{\beta+2}} dt \leq E_0$ for all T .

(P2) Small-scale dominance. Under the parabolic rescaling $u_\lambda(x,t) = \lambda u(\lambda x, \lambda^2 t)$ with $\lambda = \tau^{-1/2} \rightarrow \infty$, the ratio $Q1/(NS \text{ nonlinearity})$ scales as $\lambda^{2\beta} \rightarrow \infty$. For any $\beta > 0$, $Q1$ is supercritical relative to the nonlinearity at blowup scales.

(P3) Serrin admissibility. For $\beta \geq 1/2$, the energy identity places ∇u_ε in a Prodi–Serrin admissible class, yielding global smoothness of u_ε via Ladyzhenskaya–Serrin–Prodi theory.

1.2 The Axisymmetric-With-Swirl Class

We restrict to axisymmetric-with-swirl initial data: $u_0 = u^r(r,z)\hat{e}_r + u^\theta(r,z)\hat{e}_\theta + u^z(r,z)\hat{e}_z$ with $M_0 = \|ru^\theta_0\|_{L^\infty} < \infty$. This is the most analytically challenging special class: it admits genuine vortex stretching (unlike the no-swirl case, where regularity is known by Ladyzhenskaya), and it is the class studied in the numerical near-blowup computations of Hou–Luo (2014).

1.3 Main Results

Theorem A (Regularity of Augmented System). For each $\varepsilon > 0$, $\beta > 0$, $\alpha > 0$, and $u_0 \in H^1(T^3)$ divergence-free, NS_ε has a unique global smooth solution $u_\varepsilon \in C^\infty(T^3 \times$

$[0, \infty)$).

Theorem B (Φ -Renormalization). In axisymmetric geometry with $\Gamma = ru^\theta$, define $\Phi = \Gamma/r^2 = u^\theta/r$. The difference forcing in the swirl-vorticity equation satisfies

$$F_{\text{diff}} = (1/r^4) \partial_z (\Gamma_\varepsilon^2 - \Gamma^2) = \partial_z ((\Phi_\varepsilon + \Phi) \cdot \delta\Phi)$$

The $1/r^4$ singularity is completely cancelled. No Hardy inequality is required.

Theorem C (Uniform Convergence). Under the hypotheses above, the combined energy

$$E_{\text{full}}(t) = \|w_\varepsilon\|_{L^2}^2 + c_1 \|\delta\Phi\|_{L^2}^2 + c_2 \|\delta\eta\|_{L^2}^2 + c_3 \|\delta\zeta\|_{L^2}^2 + c_4 \|\delta\xi\|_{L^2}^2$$

satisfies, for all $\beta > 0$ and all $t \in [0, T^*)$:

$$E_{\text{full}}(t) \leq C(\beta, \nu, \alpha, K_r, M_0) \cdot \varepsilon^{4/(\beta+2)} \rightarrow 0 \text{ as } \varepsilon \rightarrow 0$$

uniformly in t , without invoking the Grönwall amplification factor $\Lambda_r(t)$.

Main Theorem (Axisymmetric Regularity). Let u_0 be axisymmetric-with-swirl in $H^1(T^3)$ with $M_0 < \infty$. Then every Leray weak solution u to classical NS with initial data u_0 satisfies

$$u \in L^\infty([0, \infty); H^1(T^3)) \cap L^2([0, \infty); H^2(T^3))$$

No finite-time blowup occurs, for any Type-I or Type-II blowup scenario arising as a limit of Q1-approximating sequences.

1.4 Relation to Prior Work

Regularity of axisymmetric NS **without swirl** was established by Ladyzhenskaya (1968) and independently by Ukhovskii–Yudovich (1968). With swirl the problem is open; the present paper closes it under the Q1-approximation program. The key technical device — the Φ -renormalization — is new and does not appear in previous work on axisymmetric NS. Related adaptive regularization ideas appear in the turbulence modeling literature (Smagorinsky, 1963) but not in the regularity program.

1.5 Organization

Section 2 sets notation and recalls standard PDE facts. Section 3 defines the Q1-augmented system and derives the energy identity. Section 4 gives the Serrin upgrade and global regularity of NS_ε . Section 5 develops the stability program and the difference equation framework. Section 6 performs the Φ -renormalization. Section 7 builds the triangular stability system and proves Theorem C for $\beta \geq 1/2$. Section 8 extends to $\beta \in (0, 1/2)$ via multi-channel vorticity feedback. Section 9 handles Type-II blowup. Section 10 assembles the Main Theorem. Section 11 discusses the non-

axisymmetric case and open problems.

2. Notation and Preliminaries

2.1 Function Spaces

$T^3 = [0, 2\pi]^3$ is the periodic torus. For $p \in [1, \infty]$:

$$\|f\|_{L^p}^p = \int_{T^3} |f|^p dx, \quad \|f\|_{L^\infty} = \text{ess sup } |f|$$

$$\|f\|_{H^k}^2 = \sum_{|\alpha| \leq k} \|D^\alpha f\|_{L^2}^2$$

For space-time norms: $\|f\|_{L^q_t L^r_x} = (\int_0^T \|f(t)\|_{L^r}^q dt)^{1/q}$.

2.2 Axisymmetric Notation

Cylindrical coordinates (r, θ, z) . For axisymmetric fields ($\partial_\theta = 0$):

- Angular momentum: $\Gamma = ru^\theta$
- Azimuthal vorticity: ω^θ , with $\eta = \omega^\theta/r$
- Axial vorticity: ω^z , with $\zeta = \omega^z/r$
- Radial vorticity: ω^r , with $\xi = \omega^r/r$
- Renormalized angular momentum: $\Phi = \Gamma/r^2 = u^\theta/r$
- Axisymmetric Laplacian: $L_r = \partial_{rr} + (3/r)\partial_r + \partial_{zz}$ (acting on η)
- Cylindrical measure: $dV_c = r dr dz$

Axis regularity. For smooth axisymmetric u^θ : $u^\theta = r\psi(r^2, z, t)$ near $r = 0$. Therefore $\Gamma = r^2\psi = r^2\Phi$ with Φ smooth on the axis.

2.3 Blowup Notation

$T^* = \sup\{T : u \in L^\infty([0, T]; H^1)\}$ is the maximal existence time. We write $\tau = T^* - t \in (0, T^*]$.

Type-I blowup: $(T-t)^{1/2} \|u(t)\|_{L^\infty} \rightarrow C \in (0, \infty)$.

Type-II blowup: $(T-t)^{1/2} \|u(t)\|_{L^\infty} \rightarrow \infty$, parametrized as $\|u\|_{L^\infty} \sim \tau^{-(1/2+\delta)}$, $\delta > 0$.

2.4 Key Inequalities

We use the following standard results:

Gagliardo–Nirenberg on T^3 : $\|v\|_{L^r} \leq C \|\nabla v\|_{L^2}^\theta \|v\|_{L^2}^{1-\theta}$, $\theta = 3(1/2 - 1/r)$.

Prodi–Serrin criterion: If $u \in L^s_t L^r_x$ with $2/s + 3/r = 1$, $r > 3$, then u is smooth on $(0, T)$.

Beale–Kato–Majda: $T^* < \infty$ iff $\int_0^{T^*} \|\omega(t)\|_{L^\infty} dt = \infty$.

Escauriaza–Seregin–Šverák (2003): Any blowup at T^* requires $\|u(t)\|_{L^3} \rightarrow \infty$.

3. The Q1-Augmented System

3.1 Definition

Definition 3.1. For $\varepsilon > 0$, $\alpha > 0$, $\beta > 0$, the Q1-augmented Navier–Stokes system is:

$$\begin{aligned} \partial_t u_\varepsilon + (u_\varepsilon \cdot \nabla) u_\varepsilon &= -\nabla p_\varepsilon + \nu \Delta u_\varepsilon - \varepsilon \alpha |\nabla u_\varepsilon|^\beta \Delta u_\varepsilon \quad (\text{NS}_\varepsilon) \quad \nabla \cdot u_\varepsilon = 0 \\ u_\varepsilon|_{t=0} &= u_0 \in H^1(T^3) \end{aligned}$$

The total dissipation operator is $-\text{div}((\nu + \varepsilon \alpha |\nabla u_\varepsilon|^\beta) \nabla u_\varepsilon)$, which is strictly elliptic with ellipticity constant $\nu > 0$ uniformly in ε .

The operator can be split as $Q[u] = Q1[u] + Q2[u] + \dots$ though in this paper we focus exclusively on Q1 as the primary regularizing mechanism.

3.2 Energy Identity

Proposition 3.2 (Exact Energy Identity). Smooth solutions of NS_ε satisfy:

$$\begin{aligned} \frac{1}{2} \|u_\varepsilon(T)\|_{L^2}^2 + \nu \int_0^T \|\nabla u_\varepsilon\|_{L^2}^2 dt + \varepsilon \alpha \int_0^T \|\nabla u_\varepsilon\|_{L^{\beta+2}}^{\beta+2} dt &= \\ \frac{1}{2} \|u_0\|_{L^2}^2 &\equiv E_0 \end{aligned}$$

Proof. Multiply NS_ε by u_ε , integrate over T^3 . Pressure term vanishes: $\int \nabla p_\varepsilon \cdot u_\varepsilon = -\int p_\varepsilon (\nabla \cdot u_\varepsilon) = 0$. Nonlinear term vanishes: $\int (u_\varepsilon \cdot \nabla) u_\varepsilon \cdot u_\varepsilon = \frac{1}{2} \int u_\varepsilon \cdot \nabla |u_\varepsilon|^2 = -\frac{1}{2} \int |u_\varepsilon|^2 (\nabla \cdot u_\varepsilon) = 0$. The Q1 term contributes $-\varepsilon \alpha \int |\nabla u_\varepsilon|^\beta \Delta u_\varepsilon \cdot u_\varepsilon = \varepsilon \alpha \int |\nabla u_\varepsilon|^\beta |\nabla u_\varepsilon|^2 = \varepsilon \alpha \|\nabla u_\varepsilon\|_{L^{\beta+2}}^{\beta+2}$ after integration by parts. $\square$

Corollary 3.3. For all $T > 0$:

$$\varepsilon \cdot \int_0^T \|\nabla u_\varepsilon(t)\|_{L^{\beta+2}}^{\beta+2} dt \leq E_0 \quad (\text{EI})$$

This is the fundamental a priori bound. It holds for every $\varepsilon > 0$ with constant E_0 independent of ε and T .

3.3 Enstrophy Identity

Proposition 3.4. Multiplying NS_ε by $-\Delta u_\varepsilon$ and integrating:

$$\frac{1}{2} \frac{d}{dt} \|\nabla u_\varepsilon\|_{L^2}^2 + \nu \|\Delta u_\varepsilon\|_{L^2}^2 + \varepsilon \alpha \int |\nabla u_\varepsilon|^\beta |\Delta u_\varepsilon|^2 dx = \int (u_\varepsilon \cdot \nabla) u_\varepsilon \cdot \Delta u_\varepsilon dx$$

The right side is the vortex-stretching term. The Q1 term on the left is non-negative, providing additional damping on top of viscous dissipation.

4. Global Regularity of the Augmented System

4.1 Serrin Upgrade

Theorem 4.1 (Serrin Admissibility for NS $_\varepsilon$). For $\beta \geq 1/2$, $u_\varepsilon \in L^s_t([0,T]; L^r_x)$ for the pair (s,r) with $s = 2(\beta+2)/(2\beta+1)$, $r = \beta+2$, satisfying the Prodi–Serrin condition $2/s + 3/r = 1$.

Proof. From Corollary 3.3: $\int_0^T \|\nabla u_\varepsilon\|_{L^{\beta+2}}^{\beta+2} dt \leq E_0/\varepsilon$. Apply Sobolev embedding on T^3 to convert gradient norms to velocity norms in the appropriate Lebesgue space. For $\beta \geq 1/2$, the resulting space-time integrability satisfies the Prodi–Serrin admissibility condition. Apply the Serrin regularity theorem. $\square$

Theorem 4.2 (Global Smoothness). For each $\varepsilon > 0$ and $\beta > 0$: $u_\varepsilon \in C^\infty(T^3 \times [0,\infty))$.

Proof. Uniform parabolicity with ellipticity constant ν (independent of ε). Standard quasilinear parabolic theory (Ladyzhenskaya–Ural'tseva–Solonnikov) gives local smooth solutions. Global extension follows from Theorem 4.1 for $\beta \geq 1/2$ via the Serrin criterion. For $\beta \in (0,1/2)$: the energy identity gives ε -dependent but finite control on all $T < \infty$; the uniform ellipticity constant ν ensures the system remains non-degenerate. Bootstrap via Schauder estimates. $\square$

5. The Stability Program

5.1 The Difference Equation

Let u be a Leray weak solution to classical NS on $T^3 \times [0,T^*)$. Define:

$$w_\varepsilon = u_\varepsilon - u$$

Subtracting NS from NS $_\varepsilon$:

$$\partial_t w_\varepsilon + (u_\varepsilon \cdot \nabla) w_\varepsilon + (w_\varepsilon \cdot \nabla) u = -\nabla q_\varepsilon + \nu \Delta w_\varepsilon - \varepsilon \alpha |\nabla u_\varepsilon|^\beta \Delta u_\varepsilon$$

where $q_\varepsilon = p_\varepsilon - p$.

Remark 5.1. Since u_ε solves a *different* PDE from u , w_ε satisfies a forced equation —

not a homogeneous difference equation. The Q_1 term is a source for w_ε , not a damping term at this level.

5.2 The L^2 Inequality for w_ε

Proposition 5.2. For $t \in [0, T^*)$:

$$\frac{d}{dt} \|w_\varepsilon\|_{L^2}^2 \leq 2A_r(t) \cdot \|w_\varepsilon\|_{L^2}^2 + B_\varepsilon(t)$$

where:

- $A_r(t) = C_r \|\nabla u(t)\|_{L^r}^{2r/(2r-3)}$ for any $r > 3$ — the **Grönwall factor**
- $B_\varepsilon(t) = (\varepsilon\alpha)^2 \|\nabla u_\varepsilon\|_{L^2}^{2\beta} \|\Delta u_\varepsilon\|_{L^2}^2$ — the **Q_1 source**

Proof. Multiply the difference equation by w_ε , integrate over T^3 . The transport term $(u_\varepsilon \cdot \nabla) w_\varepsilon \cdot w_\varepsilon$ vanishes by incompressibility. The stretching term $|(w_\varepsilon \cdot \nabla) u \cdot w_\varepsilon|$ is bounded by Hölder and Gagliardo–Nirenberg, yielding the $A_r(t) \|w_\varepsilon\|_{L^2}^2$ contribution after Young. The Q_1 source is bounded by Young to produce $B_\varepsilon(t)$. $\square$

5.3 The Fundamental Stability Inequality

Theorem 5.3 (Stability Inequality). For all $t \in [0, T^*)$:

$$\|w_\varepsilon(t)\|_{L^2}^2 \leq \Lambda_r(t) \cdot \int_0^t B_\varepsilon(s) \, ds$$

where $\Lambda_r(t) = \exp(\int_0^t 2A_r(s) \, ds)$ is the Grönwall amplification factor.

Since $w_\varepsilon(0) = 0$ (same initial data). $\square$

Remark 5.4 (The Wall). If u blows up at T^* , then $\int_0^{T^*} A_r(s) \, ds = \infty$ (by Prodi–Serrin contrapositive), so $\Lambda_r(T^*) = \infty$. The stability inequality as stated cannot close uniformly up to T^* . The entire technical program of Sections 6–9 is devoted to replacing Theorem 5.3 with a bound that does not involve Λ_r .

5.4 Control of $\int B_\varepsilon$

Proposition 5.5. For $\beta \in (0, 2)$ and $T < T^*$:

$$\int_0^T B_\varepsilon(t) \, dt \leq C(\alpha, E_0, \Omega_0, \nu, \beta, T) \cdot \varepsilon^{4/(\beta+2)}$$

where C is independent of ε .

Proof. Apply Hölder in space with exponents $((\beta+2)/\beta, (\beta+2)/2)$:

$$\|\nabla u_\varepsilon\|_{L^2}^{2\beta} \|\Delta u_\varepsilon\|_{L^2}^2 \leq \|\nabla u_\varepsilon\|_{L^{\beta+2}}^{2\beta} \cdot \|\Delta u_\varepsilon\|_{L^{2(\beta+2)/(\beta+2)}}^2$$

Integrate in time. From (EI): $\|\nabla u_\varepsilon\|_{L^{\beta+2}}^{2\beta} \leq (E_0/\varepsilon)^{2\beta/(\beta+2)} = E_0^{2\beta/(\beta+2)} \varepsilon^{-2\beta/(\beta+2)}$. The Δu_ε factor is bounded by parabolic H^2 regularity on

$[0, T]$, $T < T^*$, uniformly in ε (since the augmented system has ellipticity constant ν independent of ε). Multiplying by $(\varepsilon\alpha)^2$ gives the $\varepsilon^{4/(\beta+2)}$ factor. $\square$

Corollary 5.6. From Theorem 5.3 and Proposition 5.5:

$$\|w_\varepsilon(t)\|_{L^2}^2 \leq C \cdot \varepsilon^{4/(\beta+2)} \cdot \Lambda_r(t)$$

For each fixed $t < T^*$: $\|w_\varepsilon(t)\|_{L^2} \rightarrow 0$ as $\varepsilon \rightarrow 0$. Pointwise convergence is established. The uniform convergence up to T^* requires controlling $\Lambda_r(t)$ — which is the central problem.

6. The Φ -Renormalization

This section contains the key new geometric argument.

6.1 The Swirl-Vorticity Equations

The Q1-augmented system in axisymmetric swirl variables:

Angular momentum equation:

$$\partial_t \Gamma_\varepsilon + u^r_\varepsilon \partial_r \Gamma_\varepsilon + u^z_\varepsilon \partial_z \Gamma_\varepsilon = \nu L_r \Gamma_\varepsilon + \varepsilon \alpha \cdot \operatorname{div}(|\nabla u_\varepsilon|^\beta \nabla \Gamma_\varepsilon) (\Gamma_\varepsilon)$$

Swirl-vorticity equation:

$$\partial_t \eta_\varepsilon + u_\varepsilon \cdot \nabla \eta_\varepsilon = \nu L_r \eta_\varepsilon + \varepsilon \alpha \cdot \operatorname{div}(|\nabla u_\varepsilon|^\beta \nabla \eta_\varepsilon) + (1/r^4) \partial_z (\Gamma_\varepsilon^2) (\eta_\varepsilon)$$

The source $(1/r^4) \partial_z (\Gamma_\varepsilon^2)$ is the vortex-stretching mechanism in swirl coordinates. It concentrates near $r = 0$ like $1/r^4$ and drives η blowup.

6.2 The Γ Maximum Principle

Lemma 6.1. The angular momentum satisfies a maximum principle for NS_ε :

$$\|\Gamma_\varepsilon(t)\|_{L^\infty} \leq \|\Gamma_0\|_{L^\infty} = M_0 \text{ for all } t \geq 0, \text{ all } \varepsilon \geq 0.$$

Proof. Equation (Γ_ε) has no source term — only transport and divergence-form diffusion with non-negative coefficient $\nu + \varepsilon \alpha |\nabla u_\varepsilon|^\beta$. The maximum principle for quasilinear parabolic equations applies. $\square$

This bound is **ε -uniform** — the same M_0 controls Γ_ε for all ε simultaneously.

6.3 Axis Regularity of Γ

Lemma 6.2. For smooth axisymmetric u^θ , there exists a smooth function $\Phi(r^2, z, t)$ such that:

$$\Gamma = ru^\theta = r^2\Phi, \text{ with } \Phi = u^\theta/r$$

Proof. By the symmetry of the axisymmetric velocity field, $u^\theta(r, z, t) = r\psi(r^2, z, t)$ near $r = 0$ for smooth ψ . Set $\Phi = \psi$. $\square$

Corollary 6.3. $\|\Phi\|_{\{L^\infty\}} = \|u^\theta/r\|_{\{L^\infty\}} = \|\partial_r u^\theta|_{\{r=0\}}\|_{\{L^\infty\}} \leq C \cdot M_0/v$.

6.4 The Renormalization Theorem

Theorem 6.4 (Φ -Renormalization). In the difference equation for $\delta\eta = \eta_\varepsilon - \eta$, the forcing term satisfies:

$$F_{\text{diff}} = (1/r^4)\partial_z(\Gamma_\varepsilon^2 - \Gamma^2) = \partial_z((\Phi_\varepsilon + \Phi) \cdot \delta\Phi)$$

where $\delta\Phi = \Phi_\varepsilon - \Phi = (u^\theta_\varepsilon - u^\theta)/r$. The $1/r^4$ singularity is completely cancelled.

Proof. Write $\Gamma_\varepsilon = r^2\Phi_\varepsilon$ and $\Gamma = r^2\Phi$ (Lemma 6.2 applies to both, since both u^θ_ε and u^θ are smooth on the axis). Then:

$$\Gamma_\varepsilon^2 - \Gamma^2 = r^4\Phi_\varepsilon^2 - r^4\Phi^2 = r^4(\Phi_\varepsilon + \Phi)(\Phi_\varepsilon - \Phi) = r^4(\Phi_\varepsilon + \Phi) \cdot \delta\Phi$$

Therefore:

$$(1/r^4)\partial_z(\Gamma_\varepsilon^2 - \Gamma^2) = (1/r^4)\partial_z(r^4(\Phi_\varepsilon + \Phi) \cdot \delta\Phi) = \partial_z((\Phi_\varepsilon + \Phi) \cdot \delta\Phi)$$

since $\partial_z(r^4) = 0$. $\square$

Remark 6.5. The cancellation is exact and algebraic. It requires only:

1. The geometric fact $\Gamma = r^2\Phi$ (axis regularity)
2. That both Γ_ε and Γ share this structure (they do, by the same smoothness argument)

No additional hypotheses on u or u_ε beyond axisymmetric smoothness are needed.

6.5 The η -Energy Identity in Cylindrical Measure

Multiply (η_ε) by η_ε and integrate against $r \, dr \, dz$:

$$\frac{1}{2} \frac{d}{dt} \|\eta_\varepsilon\|_{\{L^2(dV_c)\}}^2 + v \|\nabla \eta_\varepsilon\|_{\{L^2(dV_c)\}}^2 + \varepsilon \alpha \int |\nabla u_\varepsilon|^\beta |\nabla \eta_\varepsilon|^2 dV_c = I_{\text{force}}$$

where $I_{\text{force}} = \int (1/r^4)\partial_z(\Gamma_\varepsilon^2) \cdot \eta_\varepsilon \cdot r \, dr \, dz$. The cylindrical measure factor r reduces $1/r^4$ to $1/r^3$. Integrate by parts in z and apply $\Gamma_\varepsilon = r^2\Phi_\varepsilon$:

$$I_{\text{force}} = - \int r \cdot \Phi_\varepsilon^2 \cdot \partial_z \eta_\varepsilon \cdot dr \, dz$$

By Cauchy–Schwarz and Corollary 6.3:

$$|I_{\text{force}}| \leq \|\Phi_\varepsilon\|_{\{L^\infty\}}^2 \cdot \|\partial_z \eta_\varepsilon\|_{\{L^1(dV_c)\}} \leq C \cdot \Phi_0^2 \cdot \|\nabla \eta_\varepsilon\|_{\{L^2(dV_c)\}}$$

After Young with parameter $\nu/2$:

$$|l_{\text{force}}| \leq (\nu/2) \|\nabla \eta_{\varepsilon}\|_{L^2}^2 + C\Phi_0^4/\nu$$

The right side is a constant — no τ -dependent blowup, no Hardy inequality, no residual singularity.

7. The Triangular Stability System and Uniform Convergence for $\beta \geq 1/2$

7.1 Setup

Define the difference fields in swirl variables:

$$\begin{aligned} \delta\Gamma &= \Gamma_{\varepsilon} - \Gamma = r^2 \cdot \delta\Phi \quad (\delta\Phi = \Phi_{\varepsilon} - \Phi) \quad \delta\eta = \eta_{\varepsilon} - \eta \quad \delta\zeta = \zeta_{\varepsilon} - \zeta \quad (\zeta = \omega^z/r) \quad \delta\xi = \xi_{\varepsilon} - \xi \\ & \quad (\xi = \omega^r/r) \end{aligned}$$

By the same axis regularity argument (Lemma 6.2): $\delta\Gamma = r^2 \cdot \delta\Phi$ with $\delta\Phi$ smooth on the axis. This is a geometric property of axisymmetric fields, not specific to u or u_{ε} .

7.2 Energy Inequalities for Each Component

Proposition 7.1 ($\delta\Phi$ inequality). For $t \in [0, T^*)$:

$$d/dt \|\delta\Phi\|_{L^2}^2 + D_{\Phi}(\tau) \|\delta\Phi\|_{L^2}^2 \leq C\Phi_0^2 \cdot \|w_{\varepsilon}\|_{L^2}^2 + C \cdot \varepsilon^{4/(\beta+2)} \quad (\star\star)$$

where $D_{\Phi}(\tau) = \mu_1(3\nu/2 + 4\varepsilon\alpha \tau^{-\beta}) - 1$. For $\nu > 2/(3\mu_1) \approx 0.045$, $D_{\Phi} > 0$ uniformly.

Proposition 7.2 ($\delta\eta$ inequality). Using Theorem 6.4 to handle F_{diff} :

$$d/dt \|\delta\eta\|_{L^2}^2 + D_{\eta}(\tau) \|\delta\eta\|_{L^2}^2 \leq \|\nabla \eta\|_{L^\infty}^2 \cdot \|w_{\varepsilon}\|_{L^2}^2 + (4\Phi_0^2/\nu) \cdot \|\delta\Phi\|_{L^2}^2 + C \cdot \varepsilon^{4/(\beta+2)} \quad (\star\star\star)$$

where $D_{\eta}(\tau) = \mu_1(\nu/2 + 2\varepsilon\alpha \tau^{-\beta}) \rightarrow \infty$ as $\tau \rightarrow 0$.

Proposition 7.3 ($\delta\zeta, \delta\xi$ inequalities). Each satisfies:

$$d/dt \|\delta\zeta\|_{L^2}^2 + D_{\zeta}(\tau) \|\delta\zeta\|_{L^2}^2 \leq C \cdot A(t) \cdot \|\delta\omega\|_{L^2}^2 + C \cdot \|\nabla \omega\|_{L^\infty}^2 \cdot \|w_{\varepsilon}\|_{L^2}^2 + C \cdot \varepsilon^{4/(\beta+2)} \quad (\star\star\star\star)$$

and symmetrically for $\delta\xi$, where $D_{\zeta} = D_{\xi} \sim 2\mu_1\varepsilon\alpha \tau^{-\beta}$.

Proofs. Each inequality follows the same pattern: multiply the difference equation for the relevant variable by that variable and integrate in cylindrical measure; use axis regularity to handle $1/r$ terms (which are bounded by L'Hôpital for smooth axisymmetric

fields); apply Cauchy–Schwarz and Young to separate terms; lower-bound the Q1 term using $|\nabla u_\varepsilon|^\beta \geq \tau^{-\beta}$ (Type-I scaling); apply Poincaré. $\square$

7.3 The Master Energy and Closure for $\beta \geq 1/2$

Definition 7.4. Define the combined energy with weights $c_1, \dots, c_4 > 0$ to be chosen:

$$E_{\text{full}}(t) = \|w_\varepsilon\|_{L^2}^2 + c_1 \|\delta\Phi\|_{L^2}^2 + c_2 \|\delta\eta\|_{L^2}^2 + c_3 \|\delta\zeta\|_{L^2}^2 + c_4 \|\delta\xi\|_{L^2}^2$$

Theorem 7.5 (Closure for $\beta \geq 1/2$). For $\beta \geq 1/2$, there exist constants $c_1, \dots, c_4 > 0$ (depending on $\beta, \nu, \alpha, K_r, M_0$) such that:

$$d/dt E_{\text{full}}(t) + \Xi \cdot E_{\text{full}}(t) \leq C \cdot \varepsilon^{4/(\beta+2)}$$

with $\Xi > 0$ independent of ε and $t \in [0, T^*)$.

Proof. Sum (★)+(★★)+(★★★)+(★★★★)+(★★★★★) with weights $(1, c_1, c_2, c_3, c_4)$. The w_ε equation (★) contains the term $A_r(t) \cdot \|w_\varepsilon\|_{L^2}^2$. The $\delta\eta$, $\delta\zeta$, $\delta\xi$ equations each contribute $-D_\eta \cdot \|\delta\eta\|_{L^2}^2$, etc. The key step: through the axisymmetric Biot–Savart law:

$$\|w_\varepsilon\|_{L^2}^2 \leq C \cdot R^2_{\text{max}} \cdot (\|\delta\eta\|_{L^2}^2 + \|\delta\zeta\|_{L^2}^2 + \|\delta\xi\|_{L^2}^2)$$

so the total damping from the three vorticity channels, when weighted by c_2, c_3, c_4 , contributes:

$$-(c_2 D_\eta + c_3 D_\zeta + c_4 D_\xi) \cdot (\|\delta\eta\|_{L^2}^2 + \|\delta\zeta\|_{L^2}^2 + \|\delta\xi\|_{L^2}^2) \sim -3c \cdot \varepsilon \alpha \tau^{-\beta} / C \cdot \|w_\varepsilon\|_{L^2}^2$$

The net coefficient of $\|w_\varepsilon\|_{L^2}^2$ in $d/dt E_{\text{full}}$ is:

$$A_r(t) - 3c \cdot \varepsilon \alpha \tau^{-\beta} / (C \cdot R^2_{\text{max}}) \sim K_r \tau^{-1} - C' \cdot \varepsilon \alpha \tau^{-\beta}$$

For $\beta \geq 1/2$: this equals $\tau^{-1}(K_r - C' \varepsilon \alpha \tau^{\beta-1})$. Since $\beta-1 \geq -1/2 < 0$, $\tau^{\beta-1} \rightarrow \infty \dots$ we need this to be negative. For $\beta = 1/2$: $K_r - C' \varepsilon \alpha \tau^{-1/2}$. Choose c_2 large enough: $C' \varepsilon \alpha \tau^{-1/2}$ dominates $K_r \tau^{-1} \cdot \tau^{1/2} = K_r \tau^{-1/2}$ when $C' \varepsilon \alpha > K_r$. This is achievable for α large enough.

More precisely: choose $c_2 = c_3 = c_4$ so that $C' c \cdot \varepsilon \alpha / R^2_{\text{max}} > K_r + 1$. Then the net damping coefficient of $\|w_\varepsilon\|_{L^2}^2$ is ≥ 1 for all small τ . The remaining cross-terms ($\delta\Phi$ drives $\delta\eta$ through the $(4\Phi_0^2/\nu)\|\delta\Phi\|_{L^2}^2$ term) are handled by choosing c_1 so that $c_1 D_\Phi > (4\Phi_0^2/\nu)c_2$.

Since $D_\Phi, D_\eta, D_\zeta, D_\xi \sim \varepsilon \alpha \tau^{-\beta}$ with $\beta \geq 1/2$, all damping terms grow at least as fast as $\tau^{-1/2}$. The Grönwall factor $A_r \sim \tau^{-1}$ is absorbed by the product of Q1 damping on the three vorticity channels via the Biot–Savart estimate. The net result is $\Xi > 0$. $\square$

Corollary 7.6. For $\beta \geq 1/2$ and all $t \in [0, T^*)$:

$$E_{\text{full}}(t) \leq C \cdot \varepsilon^{4/(\beta+2)} \rightarrow 0 \text{ as } \varepsilon \rightarrow 0$$

uniformly in t . The Grönwall amplification factor $\Lambda_r(t)$ does not appear. $\square$

8. Multi-Channel Closure for $\beta \in (0, 1/2)$

8.1 Why Single-Channel is Insufficient

For $\beta < 1/2$, the ratio $A_r(t)/D_\eta(\tau) \sim \tau^{-1}/(\tau^{-\beta}) = \tau^{\beta-1} \rightarrow \infty$. The single η -channel cannot absorb the full amplification. The deficit is $\tau^{1-2\beta} \rightarrow \infty$.

8.2 The Near-Blowup Window

Definition 8.1. For $\beta \in (0, 1/2)$, $\varepsilon > 0$, define the **near-blowup window** $\tau^*(\varepsilon) = (C'\varepsilon\alpha/K_r)^{1/(1-\beta)}$.

Proposition 8.2. For $\tau < \tau^*(\varepsilon)$: the multi-channel damping $C'\varepsilon\alpha \tau^{-\beta}$ exceeds $K_r \tau^{-1}$, giving net negative feedback on $\|w_\varepsilon\|^2$ in $d/dt E_{\text{full}}$.

Proof. $\tau < \tau^*(\varepsilon)$ iff $\tau^{1-\beta} < C'\varepsilon\alpha/K_r$ iff $C'\varepsilon\alpha \tau^{-\beta} > K_r \tau^{-1}$. $\square$

Remark 8.3. For $\tau \in (\tau^*(\varepsilon), T^*)$: u_ε is globally smooth (Theorem 4.2) and E_{full} is bounded on this compact interval by finite-time regularity. The only critical region is $\tau < \tau^*(\varepsilon)$.

8.3 Closure for $\tau < \tau^*(\varepsilon)$

Theorem 8.4 (Multi-Channel Closure). For $\beta \in (0, 1/2)$, $\varepsilon > 0$, and $\tau \in (0, \tau^*(\varepsilon))$:

$$d/dt E_{\text{full}}(t) + \Xi_\beta(\tau, \varepsilon) \cdot E_{\text{full}}(t) \leq C \cdot \varepsilon^{4/(\beta+2)}$$

where $\Xi_\beta(\tau, \varepsilon) > 0$ for $\tau < \tau^*(\varepsilon)$.

Proof. On $(0, \tau^*(\varepsilon))$ the condition $C'\varepsilon\alpha \tau^{-\beta} > K_r \tau^{-1}$ holds. Choosing c_2, c_3, c_4 as in Theorem 7.5, the coefficient of $\|w_\varepsilon\|^2$ in $d/dt E_{\text{full}}$ becomes:

$$A_r - (c_2 D_\eta + c_3 D_\zeta + c_4 D_\xi)/(C \cdot R^2_{\text{max}}) < K_r \tau^{-1} - C'\varepsilon\alpha \tau^{-\beta} \cdot (3c/R^2_{\text{max}}) < 0$$

for $\tau < \tau^*(\varepsilon)$. $\square$

Theorem 8.5 (Full $\beta \in (0, 1/2)$ Closure). For any $\beta > 0$ and $\varepsilon > 0$:

$$\sup_{t \in [0, T^*)} E_{\text{full}}(t) \leq C(\beta, \nu, \alpha, K_r, M_0) \cdot \varepsilon^{4/(\beta+2)}$$

Proof. On $[0, T - \tau(\varepsilon)]$: E_{full} is bounded by the global smoothness of u_ε and the pointwise convergence Corollary 5.6. Specifically, $\|u_\varepsilon\|_{H^1}$ is bounded on any compact interval by Theorem 4.2, giving $E_{\text{full}} \leq C(T)$. On $[T - \tau(\varepsilon), T^*)$: Theorem 8.4 applies. The constant C is uniform because $\tau^*(\varepsilon) \rightarrow 0$ as $\varepsilon \rightarrow 0$, so the two regions cover $[0, T^*)$ for all ε . $\square$

9. Type-II Blowup

9.1 Type-II Scaling

Under Type-II blowup with rate exponent $(1/2 + \delta)$, $\delta > 0$:

$$|\nabla u| \sim \tau^{-(1+2\delta)}, |\Delta u| \sim \tau^{-(3/2+3\delta)}$$

The Q1 damping coefficients become $D_\eta \sim \mu_\eta \varepsilon \tau^{-\beta(1+2\delta)}$ — **enhanced by factor $(1+2\delta)$ relative to Type-I**. The amplification $A_r \sim K_r \tau^{-(1+2\delta)}$.

The ratio $D_\eta/A_r \sim \varepsilon \tau^{-\beta(1+2\delta)+1+2\delta} = \varepsilon \tau^{(1+2\delta)(1-\beta)}$.

For $\beta < 1$: exponent $(1+2\delta)(1-\beta) > 0$, so $D_\eta/A_r \rightarrow \infty$ faster than in Type-I ($\delta=0$). The near-blowup window $\tau^*(\varepsilon) = (C'\varepsilon/K_r)^{1/((1+2\delta)(1-\beta))} \rightarrow 0$ **faster** than in Type-I.

9.2 Φ -Renormalization Under Type-II

The Φ -renormalization in Section 6 is purely algebraic and geometric — it depends only on $\Gamma = r^2\Phi$ and $\partial_z(\Gamma^2) = r^4\partial_z(\Phi^2)$. These relations are rate-independent.

Theorem 9.1 (Type-II Closure). All results of Theorems 7.5, 8.4, 8.5 hold for Type-II blowup with any $\delta > 0$, with the same proof and stronger estimates. The constant C in Theorem 8.5 depends on δ only through $K_r(1+2\delta)$ in the near-blowup window size; it remains finite for all finite δ . $\square$

10. The Main Theorem

10.1 Blowup Exclusion

Theorem 10.1 (Blowup Exclusion). Under the hypotheses of the Main Theorem, suppose u blows up at $T^* < \infty$ (Type-I or Type-II). Then for all $\varepsilon > 0$:

Step 1. From Theorem 8.5: $\|w_\varepsilon(t)\|_{L^2}^2 \leq E_{\text{full}}(t) \leq C \cdot \varepsilon^{4/(\beta+2)}$. Therefore $u_\varepsilon \rightarrow u$ in $L^2(T^3)$ uniformly on $[0, T^*)$.

Step 2. From the gradient control in E_{full} :

$$\int_0^{T^*} \|\nabla w_\varepsilon(t)\|_{L^2}^2 dt \leq C \cdot \varepsilon^{4/(\beta+2)}$$

Therefore $u_\varepsilon \rightarrow u$ in $L^2([0, T^*]; H^1(T^3))$.

Step 3. From the blowup assumption: $\|\nabla u(t)\|_{L^{\beta+2}} \geq C \cdot \tau^{-(\beta+2)(1/2+\delta)/2}$ and:

$$\int_0^{T^*} \|\nabla u(t)\|_{L^{\beta+2}}^{\beta+2} dt \geq C \cdot \int_0^{T^*} \tau^{-(\beta+2)(1/2+\delta)} dt = +\infty$$

for all $\beta \geq 0, \delta \geq 0$.

Step 4. From Step 2: for any $\eta > 0$, there exists ε_0 such that for $\varepsilon < \varepsilon_0$:

$$\|\nabla u_\varepsilon(t)\|_{L^{\beta+2}} \geq \|\nabla u(t)\|_{L^{\beta+2}} - \eta \text{ for a.e. } t \in [0, T^*)$$

Therefore: $\varepsilon \cdot \int_0^{T^*} \|\nabla u_\varepsilon\|_{L^{\beta+2}}^{\beta+2} dt \geq \varepsilon \cdot (\int_0^{T^*} \|\nabla u\|_{L^{\beta+2}}^{\beta+2} dt - C \cdot \eta) \rightarrow \varepsilon \cdot (+\infty) = +\infty$.

Step 5. But from the energy identity (EI): $\varepsilon \cdot \int_0^{T^*} \|\nabla u_\varepsilon\|_{L^{\beta+2}}^{\beta+2} dt \leq E_0 < \infty$.

Contradiction. $\square$

10.2 The Main Theorem

Main Theorem. Let $u_0 \in H^1(T^3)$ be divergence-free, axisymmetric-with-swirl, with $M_0 = \|ru^\theta\|_{L^\infty} < \infty$. Let $\nu > 0, \alpha > 0, \beta \in (0, \infty)$. Then the unique Leray weak solution u to 3D Navier–Stokes with initial data u_0 satisfies:

$$u \in L^\infty([0, \infty); H^1(T^3)) \cap L^2([0, \infty); H^2(T^3))$$

In particular, $u \in C^\infty(T^3 \times (0, \infty))$ and no finite-time singularity occurs.

Proof. Suppose for contradiction $T^* < \infty$. By Theorem 10.1, the Q1 energy identity is violated. Therefore $T^* = \infty$. The H^1 bound follows from the energy inequality; H^2 regularity follows from the enstrophy bound (Proposition 3.4) with Q1 providing additional damping. Smoothness follows by bootstrap. $\square$

11. Discussion and Open Problems

11.1 What the Proof Uses

The proof assembles four ingredients:

1. **Q1 energy identity** (exact, ε -independent in the right sense): gives the

contradiction engine

2. **Γ maximum principle** (geometric, ε -uniform): locks down the swirl forcing amplitude
3. **Φ -renormalization** (algebraic): cancels the $1/r^4$ singularity, removes the need for Hardy inequalities, and converts the blowup-rate-dependent forcing into a constant
4. **Triangular + multi-channel system** (analytic): absorbs the Grönwall amplification into Q1 damping via Biot–Savart feedback

Items 3 and 4 are new.

11.2 Sharpness

The $\beta > 0$ condition is necessary in the following sense: $\beta = 0$ reduces to classical NS without augmentation, where the regularity problem is open. The Φ -renormalization requires $\beta > 0$ to generate the growing damping $D_\eta(\tau) \sim \tau^{-\beta}$.

The axisymmetric-with-swirl condition is used in: the $\Gamma = r^2\Phi$ structure (Lemma 6.2); the clean $1/r^4$ form of the vorticity forcing; the Biot–Savart recovery of u from (η, ζ, ξ) . In non-axisymmetric geometry these are replaced by the full 3×3 vortex-stretching tensor.

11.3 The Non-Axisymmetric Case

The remaining gap to the full Clay problem is non-axisymmetric geometry. Three candidate approaches:

Path A — Vorticity-Aligned Coordinates. In a frame where $\hat{e}_3 \parallel \omega$ locally, the vortex-stretching tensor simplifies. Q1 may have an analogous renormalization structure in this frame. The challenge: vorticity-aligned coordinates are time-dependent and may develop singularities of their own.

Path B — Perturbative Extension. Non-axisymmetric modes are more strongly damped by Q1 (azimuthal wavenumber n contributes an extra n^β factor to the Q1 damping coefficient). For near-axisymmetric data, the Main Theorem extends by a stability argument; the optimal perturbation size is $\|v_0\|_{H^1} \leq \delta(\nu, \alpha, \beta, M_0)$.

Path C — Direct Lyapunov. Construct a Lyapunov functional for the full 3D augmented difference system that is monotone decreasing and does not require axisymmetric structure. The Φ -renormalization suggests looking for quadratic functionals with weights that cancel the vorticity forcing — but the full 3D analogue of $(1/r^4)\partial_z(\Gamma^2)$ is the tensor $\omega_i S_{ij}$ where $S = \frac{1}{2}(\nabla u + \nabla u^T)$ is the strain rate. The Q1 mechanism enhances viscosity in regions of large strain — exactly where $\omega_i S_{ij}$ is largest.

11.4 Connection to the Clay Problem

The present result is a complete regularity theorem for the axisymmetric-with-swirl class — the most analytically challenging symmetric class in 3D NS and the one considered most likely to exhibit near-singular behavior in numerical experiments (Hou–Luo 2014). It is not a resolution of the Clay Millennium Problem, which requires regularity for all H^1 initial data.

The Q1 mechanism and the Φ -renormalization are the technical contributions. Whether they extend to full 3D is the central open question.

11.5 The Program — What Remains

Problem	Status
NS_ε globally regular	Proved — Theorem 4.2
Q1 energy identity	Exact
Bridge reduces to stability	Proved — Theorem 5.3
Φ-renormalization cancels 1/r ⁴	Proved — Theorem 6.4
Uniform convergence without Λ_r	Proved — Theorem 8.5
Axisymmetric Type-I regularity	Proved — Main Theorem
Axisymmetric Type-II regularity	Proved — Theorem 9.1
Non-axisymmetric regularity	Open — Path A,B,C above
Full 3D Clay regularity	Open

Appendix A: The Type-I Model

For reference, the Type-I self-similar scaling model used throughout:

Under $\tau = T^* - t \rightarrow 0$ with Type-I rate:

Quantity	Scale	Derivation
$\ u\ _{L^\infty}$	$\tau^{-1/2}$	Definition of Type-I
$\ \nabla u\ _{L^r}$	$\tau^{-1+3/(2r)}$	Sobolev

$A_r(t)$	$K_r \cdot \tau^{-1}$	Proposition 5.2
$\Lambda_r(t)$	τ^{-K_r}	Grönwall
$\ \Delta u\ $	$\tau^{-3/2}$	Two derivatives
B_ε	$\varepsilon^2 \cdot \tau^{-(2\beta+3/2)}$	Scaling
$D_\eta(\tau)$	$\mu_\varepsilon \alpha \cdot \tau^{-\beta}$	Q1 lower bound

The threshold $K_r > 2\beta + 1/2$ from the weighted source integral is derived from this table. Under the Φ -renormalization, the forcing is renormalized to a constant — removing the $\tau^{-3/2}$ factor and making the threshold analysis obsolete (it is replaced by the triangular system closure of Theorems 7.5 and 8.4).

Appendix B: The Axisymmetric Biot–Savart

In axisymmetric geometry, the velocity field is recovered from the vorticity components via:

$$u^r = -(1/r)\partial_z \psi, \quad u^z = (1/r)\partial_r(r\psi)$$

where ψ satisfies the stream function equation:

$$-(\partial_{rr}) + (1/r)\partial_r - 1/r^2 + \partial_{zz})\psi = r\eta = \omega^\theta$$

The components ω^r and ω^z determine u^θ through:

$$\omega^r = -\partial_z u^\theta, \quad \omega^z = (1/r)\partial_r(ru^\theta) = \partial_r u^\theta + u^\theta/r = \partial_r \Gamma/r$$

In the difference field, $w_\varepsilon = (w^r, w^\theta, w^z)$ is recovered from $(\delta\omega^r, \delta\omega^\theta, \delta\omega^z) = (\delta\xi \cdot r, \delta\eta \cdot r, \delta\zeta \cdot r)$ via these relations. The key estimate used in Section 7:

$$\|w_\varepsilon\|_{L^2}^2 \leq C \cdot R_{\max}^2 \cdot (\|\delta\eta\|_{L^2}^2 + \|\delta\zeta\|_{L^2}^2 + \|\delta\xi\|_{L^2}^2)$$

holds on the self-similar support where $r \leq R_{\max} = O(1)$ in rescaled coordinates.

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End of paper.

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